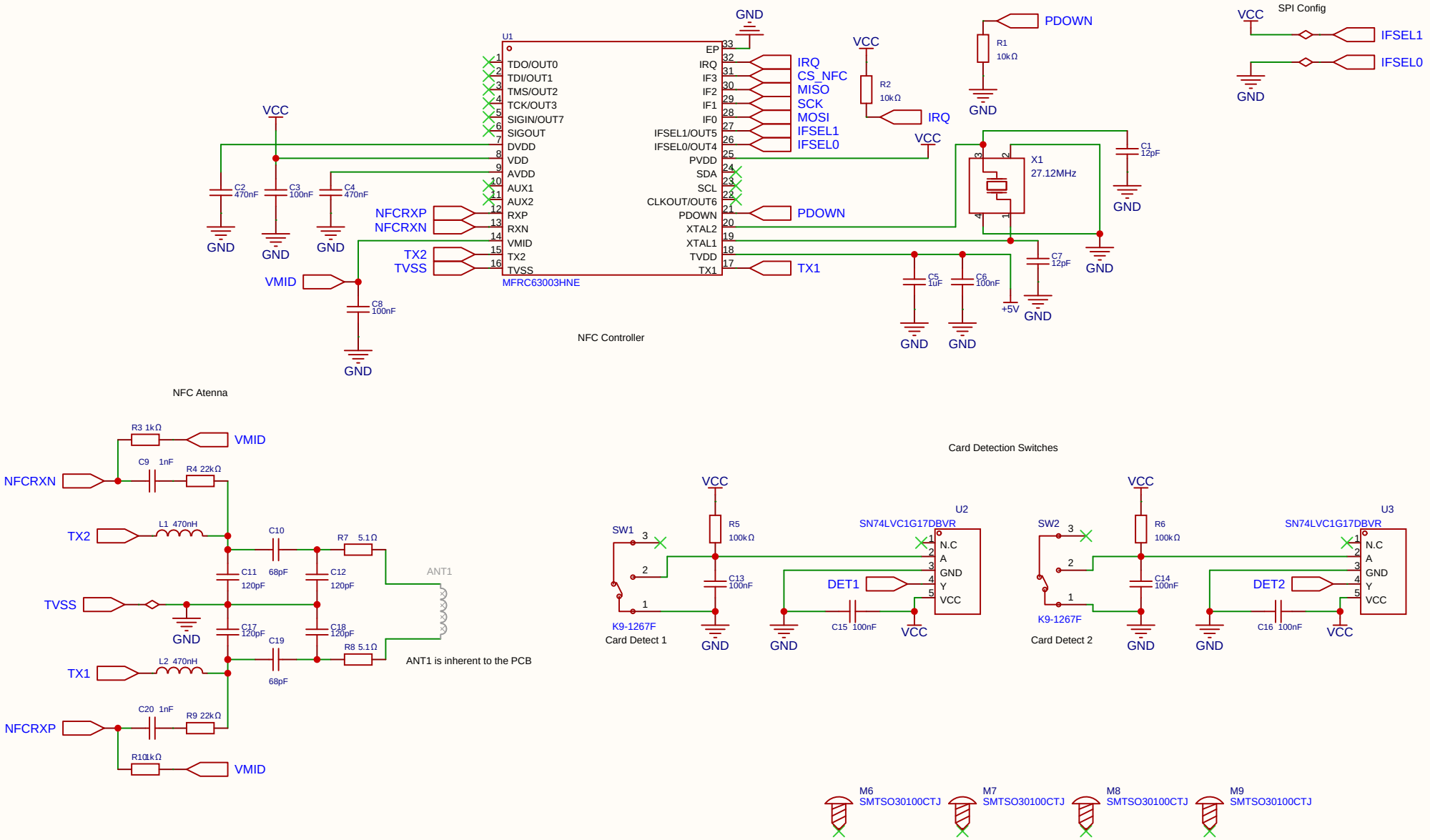
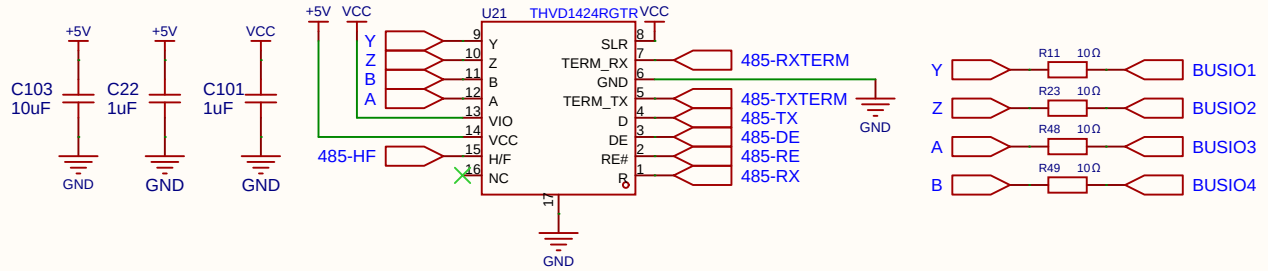


Access Control Core

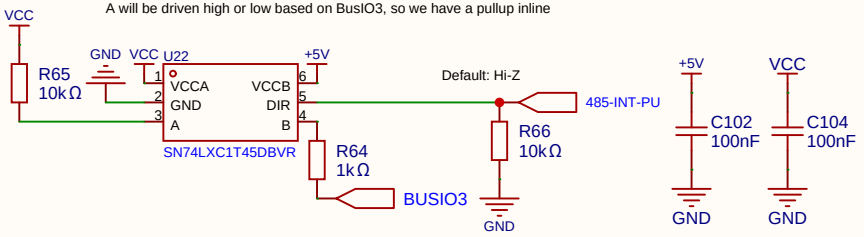
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V3.1.0 NFC Card Reader Core	NFC Reader	1 / 10

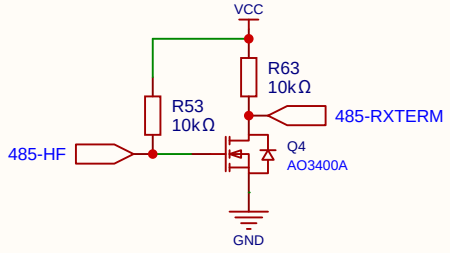
RS-485 Transceiver
Can operate in half- or full-duplex mode



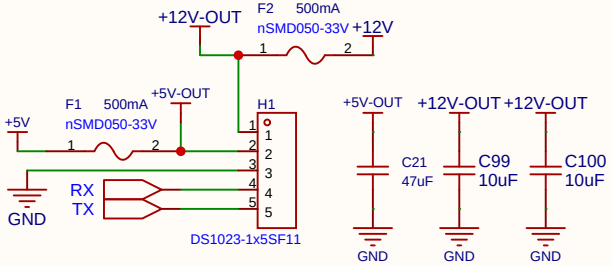
Half-Duplex Interrupt Pullup
In half-duplex mode, there is a shared interrupt on IO3
Pulled up by the Core to 5v
Read by normal bus interface
We re-use a bidirectional level shifter to achieve the High to Hi-Z outputs
When DIR is high, A->B, A is high so it drives B to +5V
Whe DIR is low, B->A, B goes Hi-Z
A will be driven high or low based on BusIO3, so we have a pullup inline



HF drives RXTERM to save GPIO:
If HF is low, we are in full duplex mode, RXTERM should be high.
If HF is high, we are either in half duplex mode, or not using RS485
Either way, RXTERM should be low
HF is externally pulled high to default half duplex, turn off RXTERM

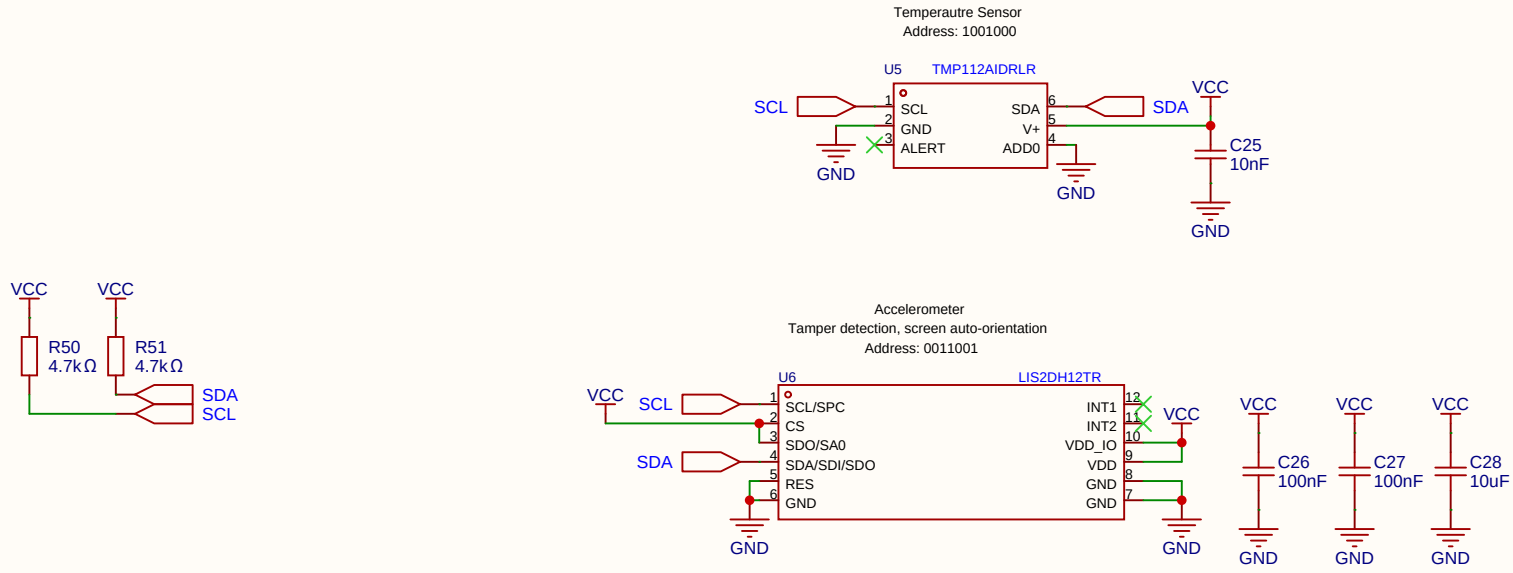


UART Header
Debug or generic expansion
Used mostly for screens



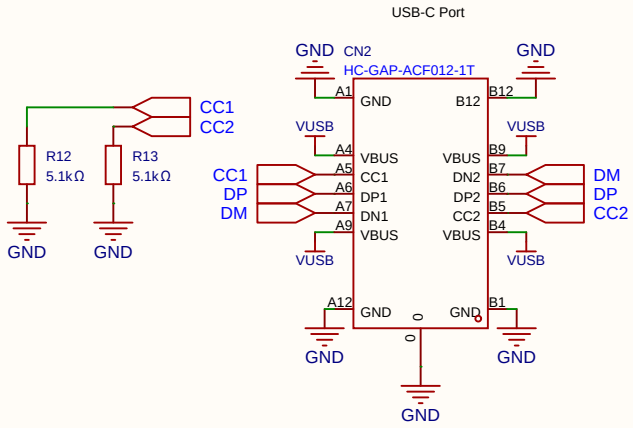
Access Control Core

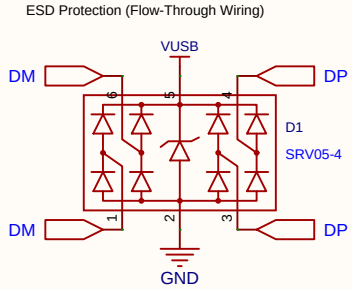
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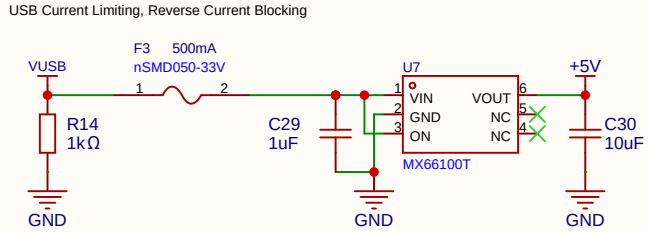


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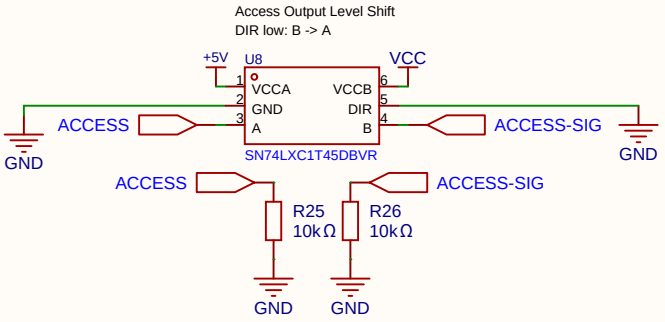


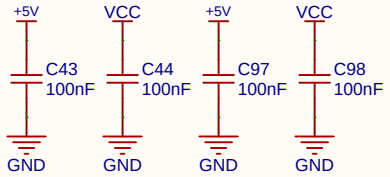


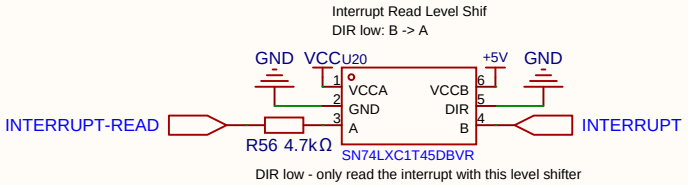


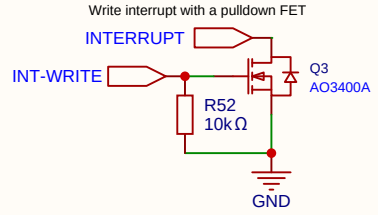
Access Control Core

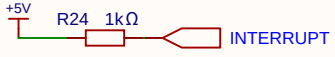
MakeACS | CERN-OHL-S 2.0 | www.github.com/MakeACS

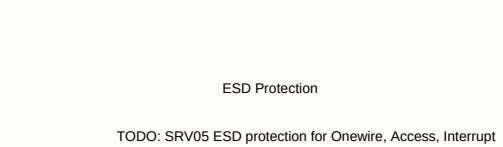


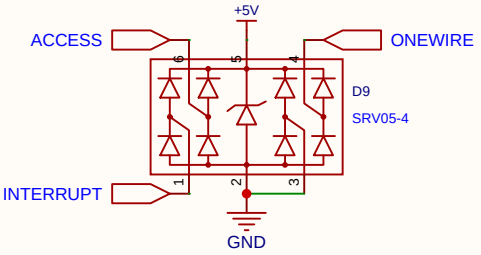


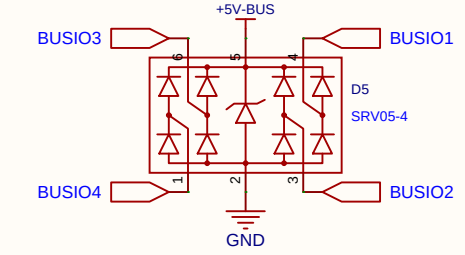


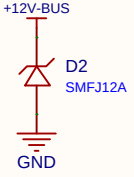


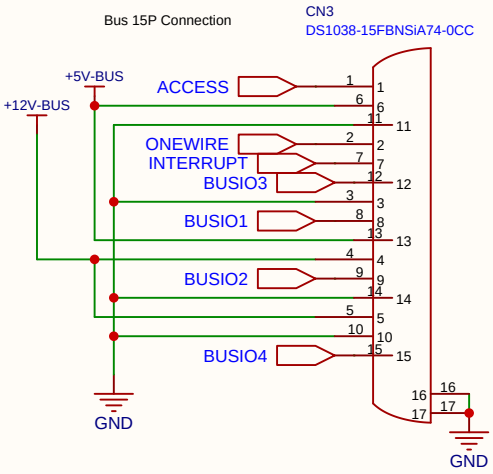


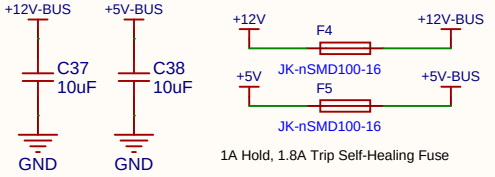


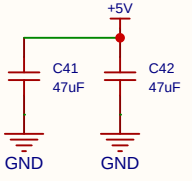


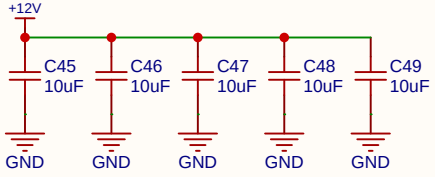










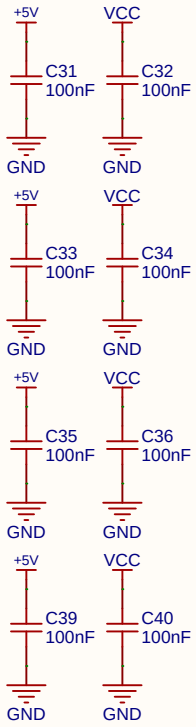


Access Control Core

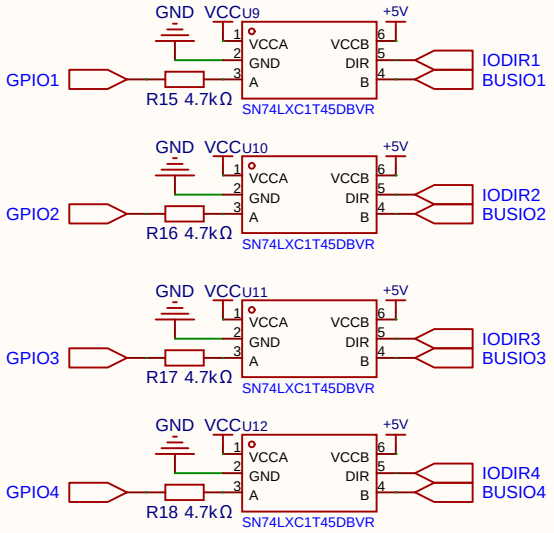
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V3.1.0 NFC Card Reader Core	Bus Connection	5 / 10

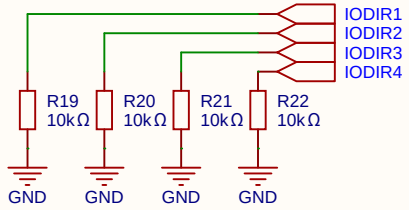
Level Shifting Capacitors



Bus GPIO Transceivers
When DIR is High, A goes to B
DIR referenced to VCCA
New in V3.1.0: Schmitt Trigger Inputs



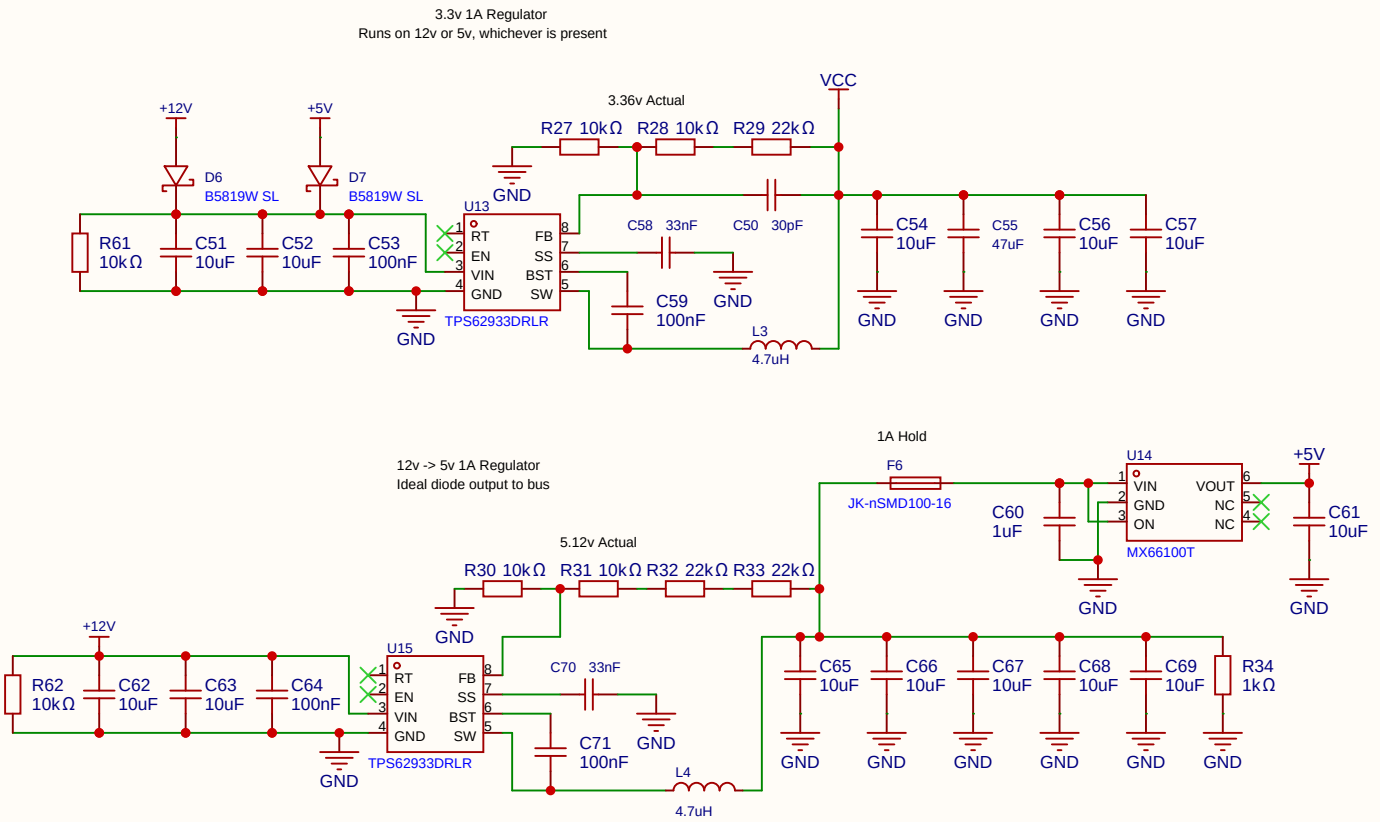
Default: Bus GPIO are inputs



Access Control Core

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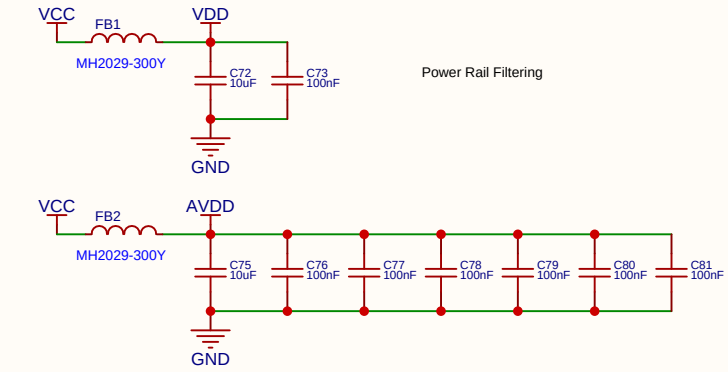
V3.1.0 NFC Card Reader Core	Bus GPIO	6 / 10

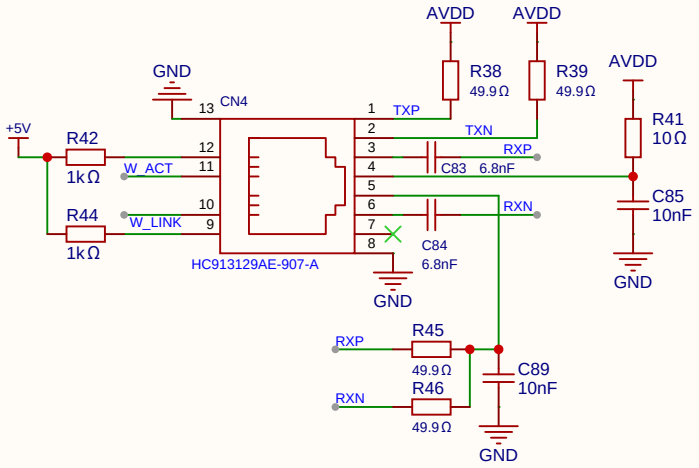


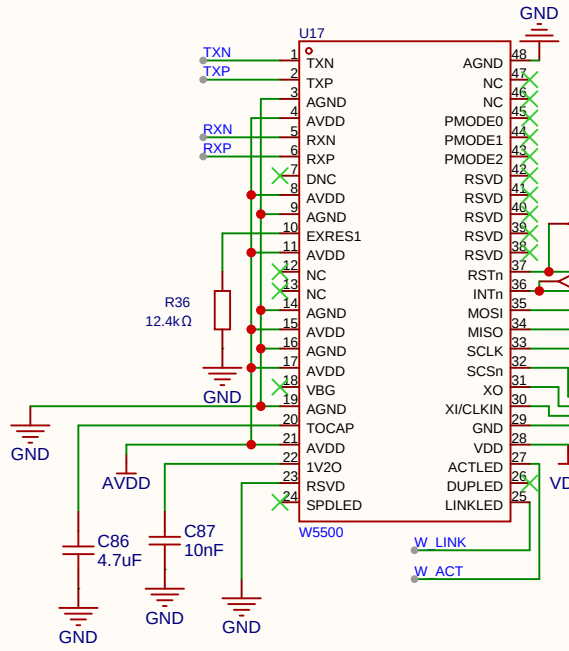
Access Control Core

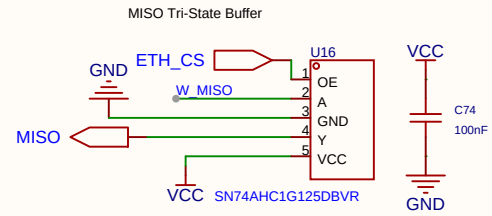
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V3.1.0 NFC Card Reader Core	Regulators	7 / 10





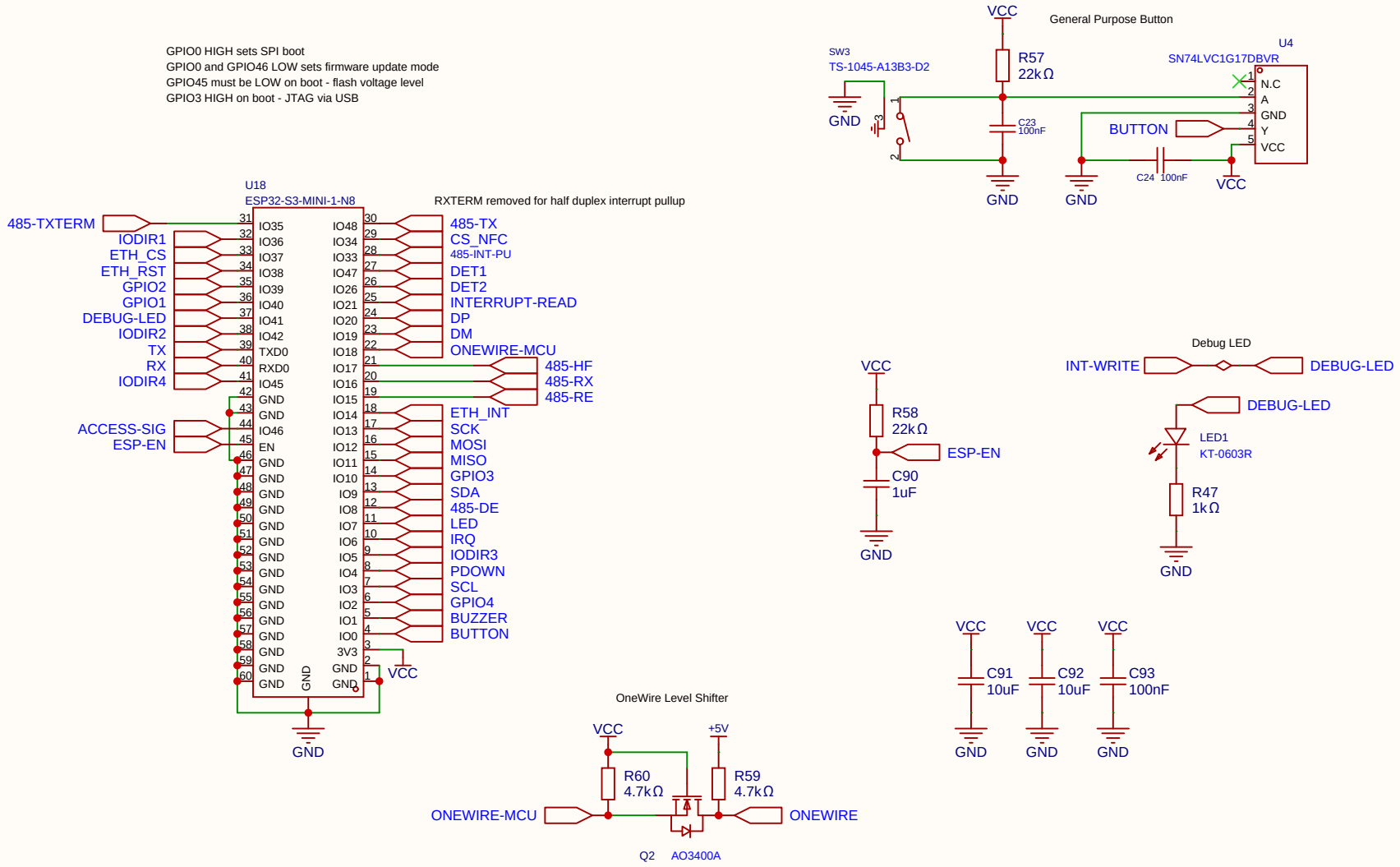




Access Control Core

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V3.1.0 NFC Card Reader Core	Ethernet	8 / 10

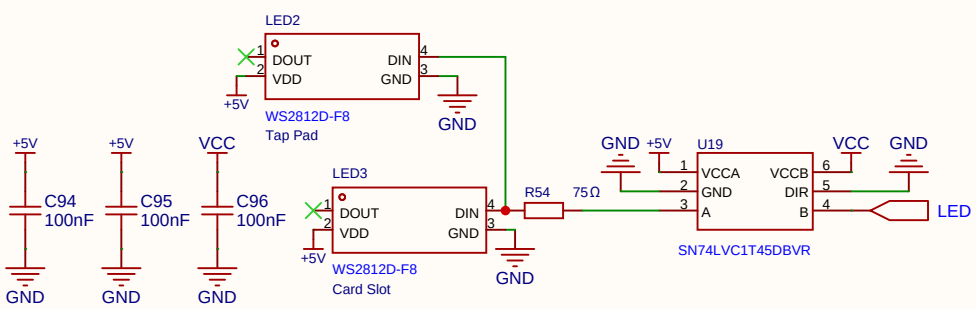


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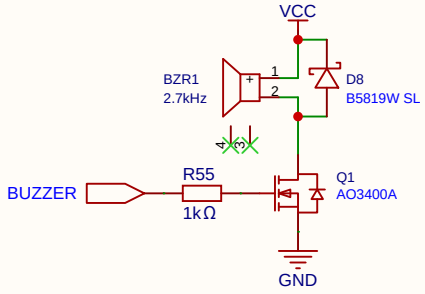
The Card Slot hardware can be easily repurposed as hardware for a Tap Pad
To convert from a card slot to a tap pad:
  Move RGB LED to LED2 position
  Replace side-fire buzzer with top-fire

RGB Status LED



By rotating the LED, it can fire upwards as a tap pad or forwards as a card slot reader

Passive Buzzer



Side Firing: MLT-8540 (C95300)
Top Firing: MLT-8540H (C95298)

A forward-firing or side-firing speaker can be used depending on the configuration.

This improves sound perception, since it will fire directly towards the user.

Access Control Core

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V3.1.0 NFC Card Reader Core	User Interface	10 / 10